

Introduction to Bayesian Statistics

Part 1 Statistical modeling

Benjamin Rosenbaum

iDiv 2026



About me

- Researcher & Statistical Consultant
- Quantitative Ecologist
- Started out as a mathematician
- Main research interests:
 - Statistical methods for process-based models
 - Population & community dynamics
 - Species interactions, functional responses



iDiv

German Centre for Integrative Biodiversity Research (iDiv)
Halle-Jena-Leipzig



EcoNetLab

Statistical support

I offer assistance for iDiv-related researchers:

- Choosing the right methods for your research topic
- Implementing statistical models in R
- Analyzing model output
- Formulating statistical methods & results in manuscripts
- Resolving statistical issues in the peer review process



Target audience

- You have used **R** before
- You have just a little bit of stats experience

Or:

- You have some stats experience, but not in Bayesian stats

Or:

- You have some Bayesian stats experience, but not with the *brms* package



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Course goals

- Building blocks of statistics modeling
Think in terms of models, not tests
 - Revision of classical models:
Learn something useful even if you want to stick to frequentist stats.
 - Basic understanding of Bayesian statistics
 - Write code with the *brms* package
 - Interpret model output & statistical inference
- Analyze your own datasets



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Contents

1. Statistical modeling
2. Bayesian principles
3. Prior and posterior distributions
4. Linear models
5. Generalized linear models
6. Mixed effects models
7. Stan introduction
8. Conclusions + questions

→ Every lesson includes a **lecture** and a **practical** part.

There are small exercises for self-study, solutions in github

This lecture

Review: probability distributions

What is a statistical model?

Probability and the likelihood function

Maximum likelihood estimation
(as preparation for Bayesian statistics)

Review: Probability distributions

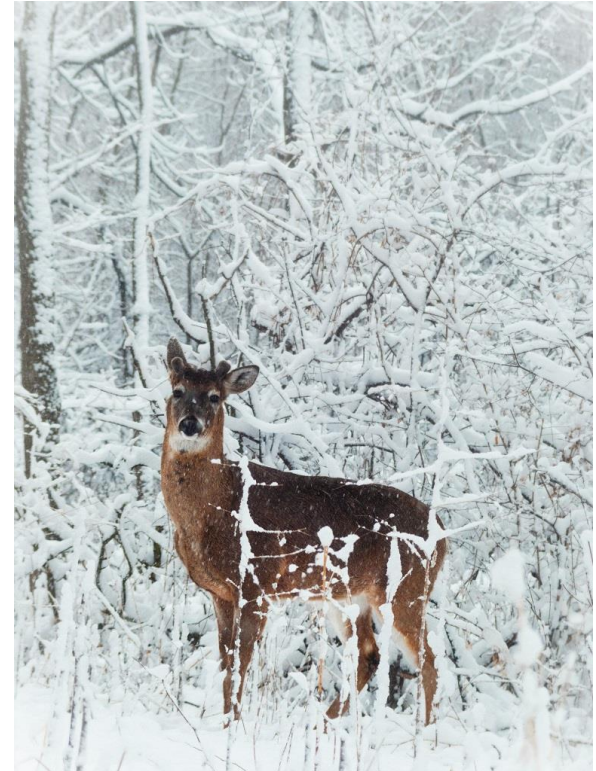
Discrete distribution

- **Example:** number of individuals from a population of $N = 10$ that survive the winter
- y **discrete** and **bounded** variable with outcomes $0, 1, 2, \dots, 10$
- Average survival probability $\theta = 0.6$ (60%)
- Binomial distribution: $y \sim \text{Binomial}(N, \theta)$

random
variable

„distributed as“

parameters:
size N
probability θ

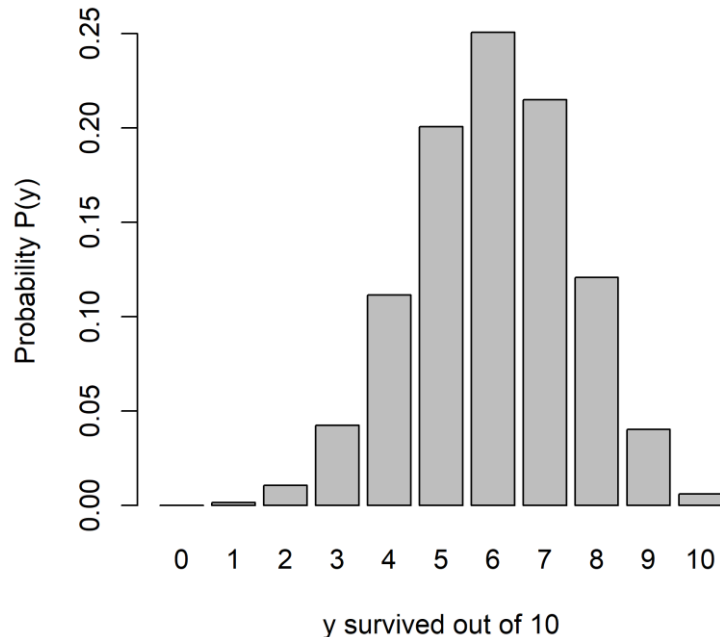


Discrete distribution

- Binomial distribution: $y \sim \text{Binomial}(N, \theta)$
- Probability function $P(y|\theta) = \binom{N}{y} \theta^y (1 - \theta)^{N-y}$
calculates **probability** of each possible outcome
for a fixed set of parameters ($N = 10, \theta = 0.6$)
- No need to memorize the equation. Use R:

```
> p = dbinom(y, size=10, prob=0.6)
```
- Draw random samples from this distribution

```
> y = rbinom(1, size=10, prob=0.6)
```



Discrete distribution

- Probabilities always sum up to 1:

$$P(y = 0) + P(y = 1) + \dots + P(y = 10) = 1$$

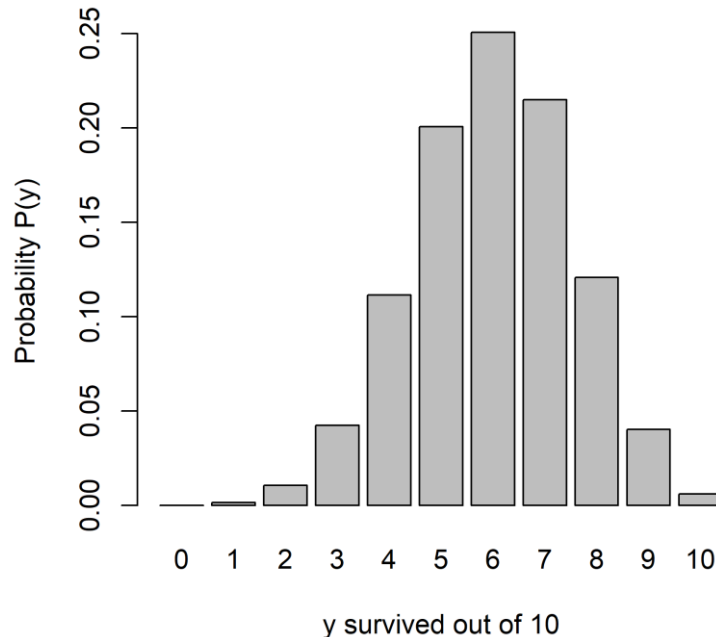
- Mean $\mu = N \cdot p = 0.6 \cdot 10 = 6$
(average outcome if experiment is repeated often)

- Compute probabilities of **events**, for example

$$P(y = 6) = 0.251$$

$$P(y \geq 6) = P(y = 6) + \dots + P(y = 10) = 0.633$$

$$P(4 \leq y \leq 8) = P(y = 4) + \dots + P(y = 8) = 0.899$$



Discrete distribution

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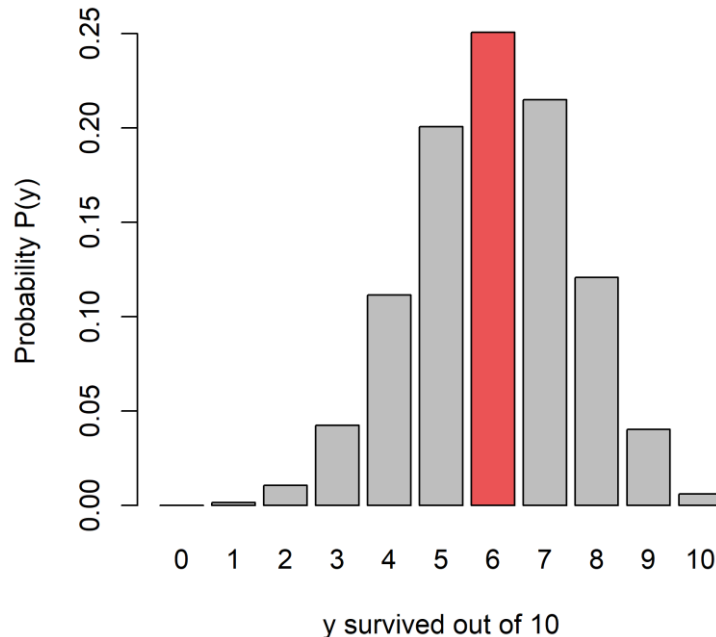
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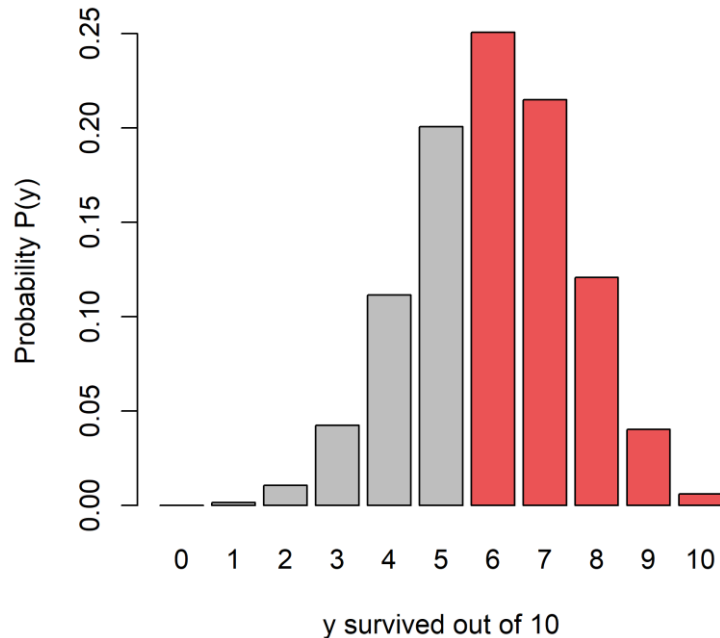
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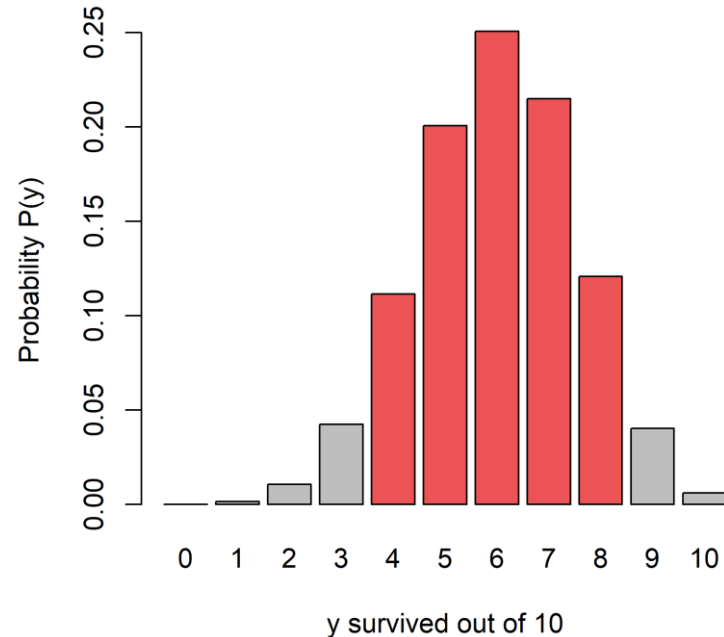
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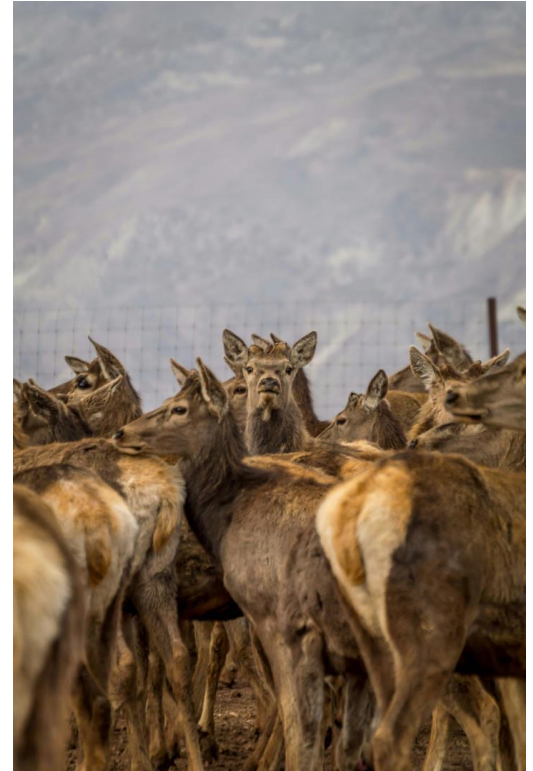
Continuous distribution

- **Example:** body mass of adult deer
- y can take any value (continuous)
- Average body mass $\mu = 100 [kg]$
- Standard deviation $\sigma = 10$ (spread)
- Normal distribution: $y \sim \text{Normal}(\mu, \sigma)$

random
variable

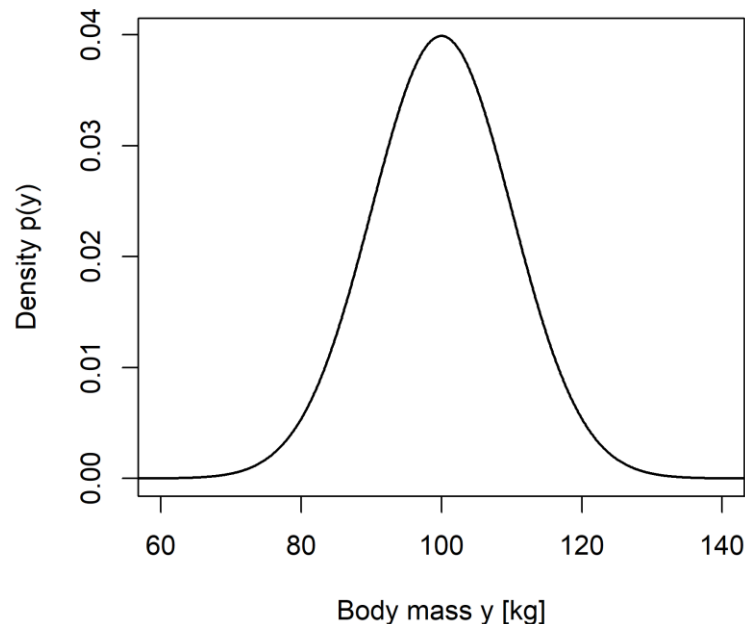
„distributed as“

parameters:
mean μ
standard deviation σ



Continuous distribution

- Normal distribution: $y \sim \text{Normal}(\mu, \sigma)$
- $p(y|\mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-\mu)^2}{2\sigma^2}}$ is the **probability density function** of each possible outcome y for a fixed set of parameters ($\mu = 100, \sigma = 10$)
- Mean μ and standard deviation σ
(average outcome if experiment is repeated often)

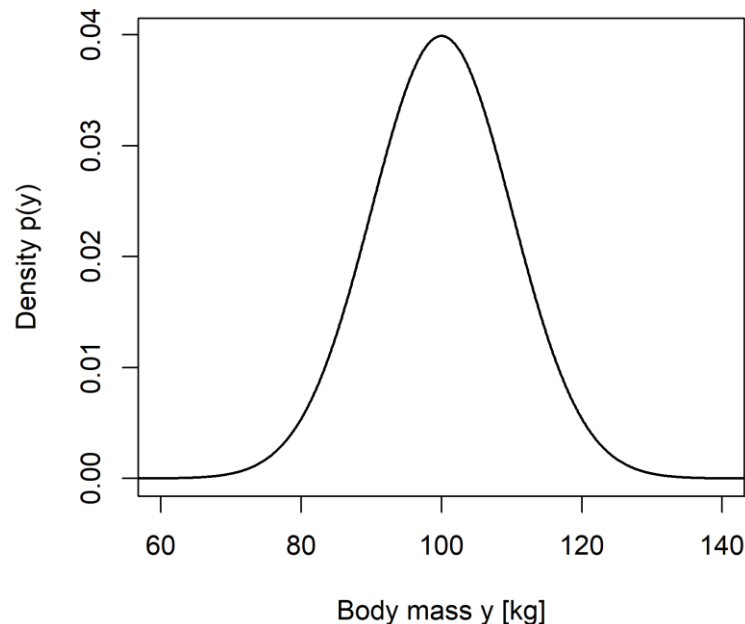


Continuous distribution

- Normal distribution: $y \sim \text{Normal}(\mu, \sigma)$
- $p(y = 95.0 \mid \mu, \sigma)$ is **not** the probability for $y = 95.0$
For continuous distributions, prob. of an exact value is zero!
(see next slide)
- No need to memorize the equation. Use R:

```
> p = dnorm(y, mean=100, sd=10)
```
- Draw random samples from this distribution

```
> y = rnorm(1, mean=100, sd=10)
```



Continuous distribution

- Probabilities always integrate to 1 (area under the curve):

$$\int p(y | \mu, \sigma) dy = 1 \text{ for any } \mu, \sigma$$

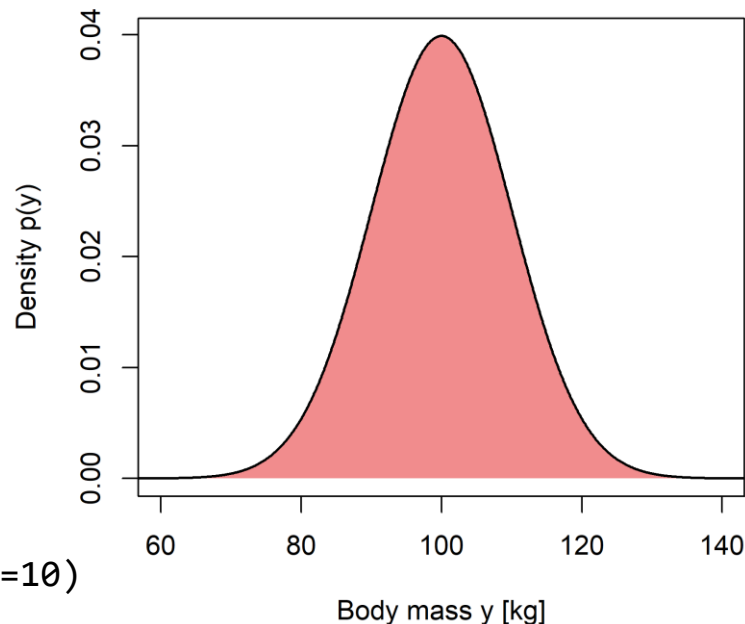
- Compute probabilities of an **interval**, for example

$$P(y \leq 110) = \int_{-\infty}^{110} p(y | 100, 10) dy = 0.841$$

```
> pnorm(110, mean=100, sd=10)
```

- $P(90 \leq y \leq 110) = \int_{90}^{110} p(y | 100, 10) dy = 0.682$

```
> pnorm(110, mean=100, sd=10) - pnorm(90, mean=100, sd=10)
```



Continuous distribution

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$$\int p(y | \mu, \sigma) dy = 1 \text{ for any } \mu, \sigma$$

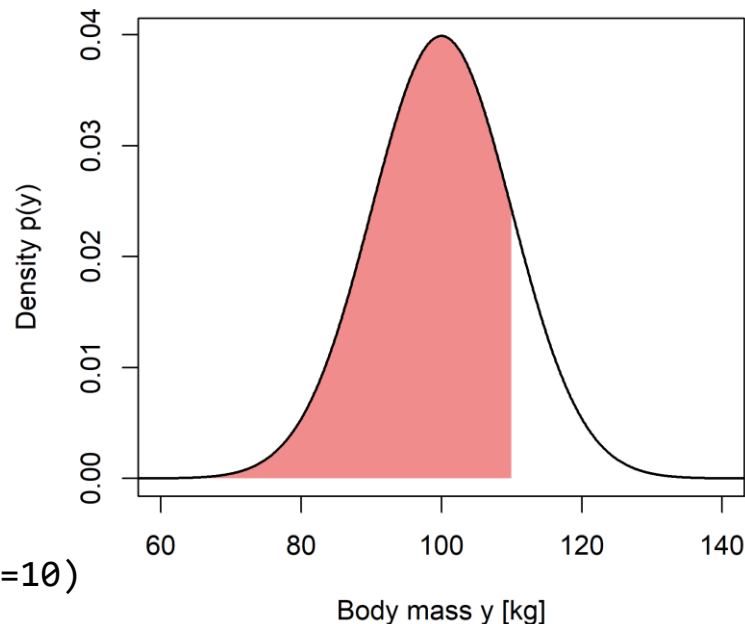
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Continuous distribution

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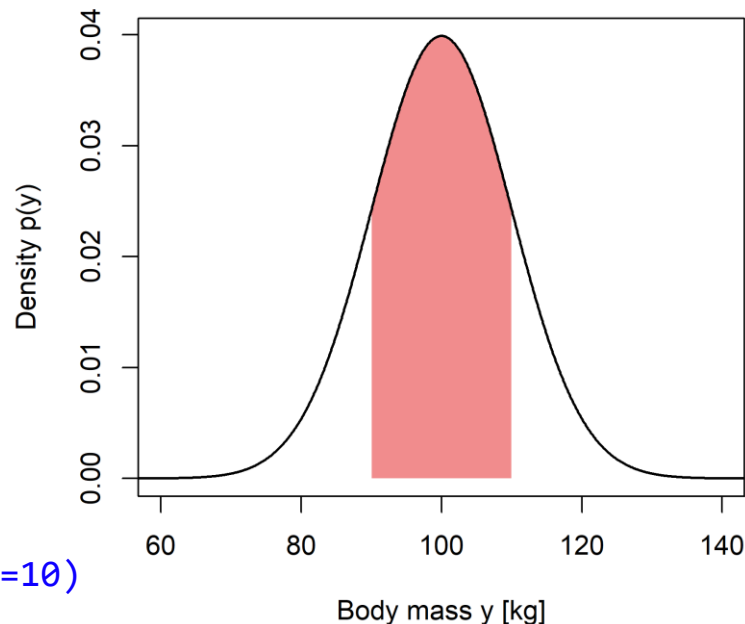
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```



Summary: distributions

A random variable y has a distribution

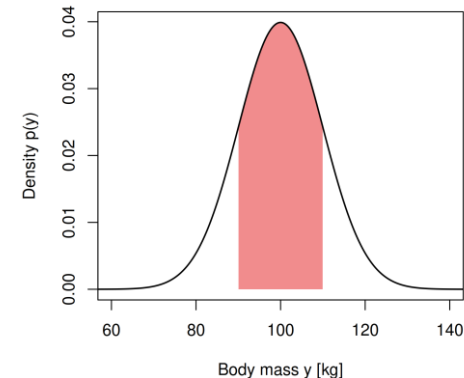
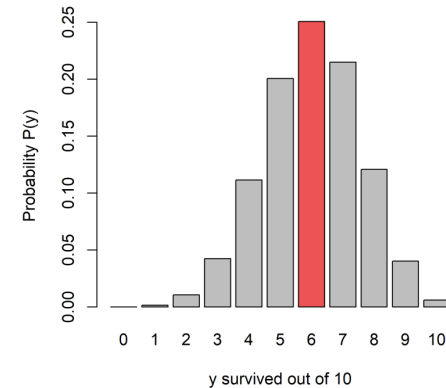
→ It has a known function to calculate probabilities

Discrete variable: „probability mass function“

- Assigns actual probability to every single value
- Probabilities sum up to 1

Continuous variable: „probability density function“

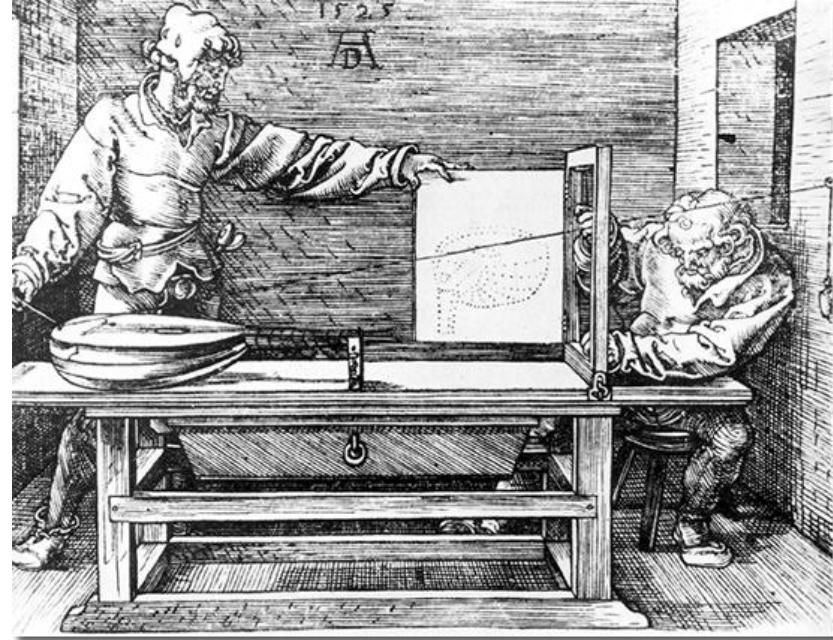
- Assigns probability density to every single value
- Actual probability of an interval = integral over density
- Density integrates to 1



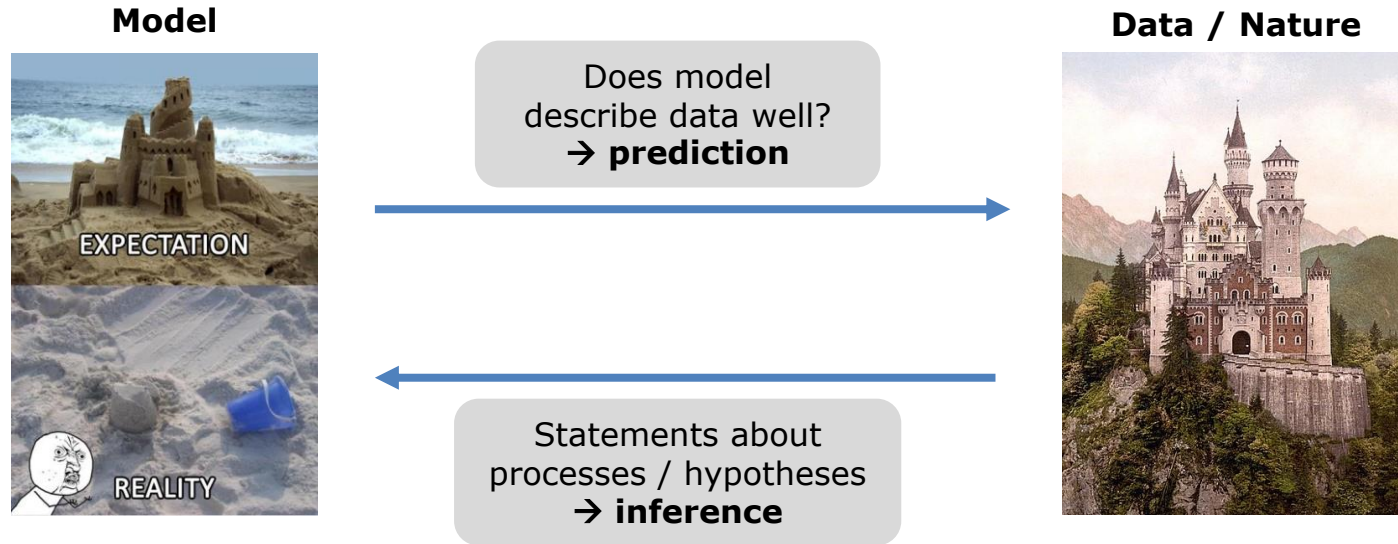
Statistical modeling

Why we need models

- Nature is **complex**. We need to simplify!
 - Models are (mathematical) **abstractions** from nature.
 - Explain **patterns** observed in nature
(associations, trends, differences, ...)
 - Make **quantitative** statements.
- Models can make sense out of your data!

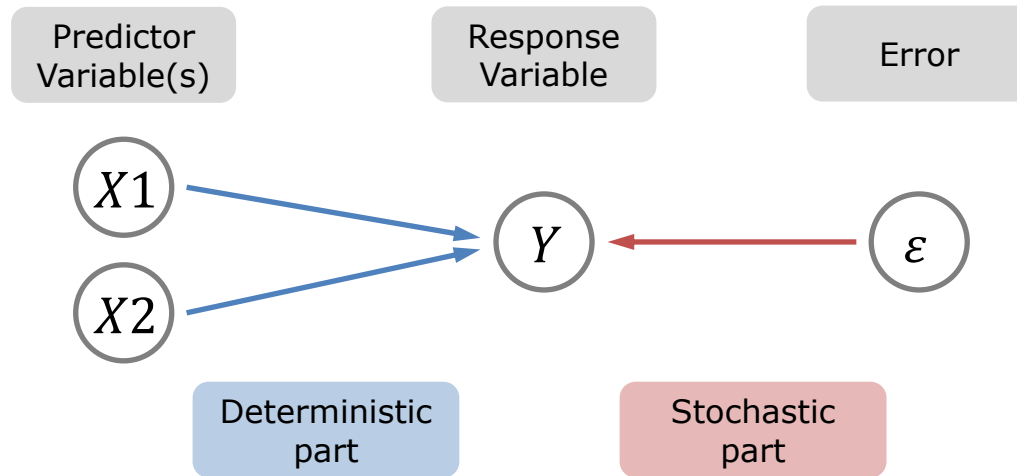


Prediction and inference



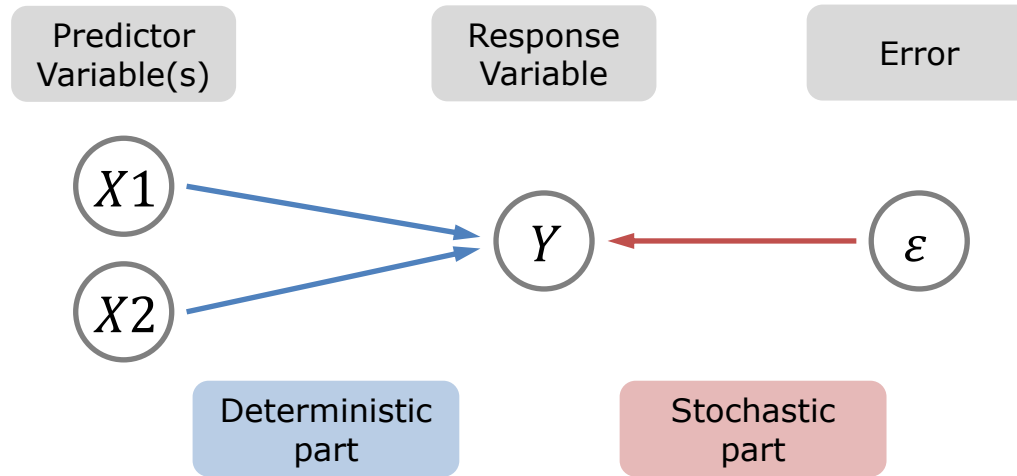
- Bring model **predictions** in correspondance with observed data
Model fitting: estimate model parameters
Model selection: choose between different models
- **Inference:** What does the data tell me about the model (e.g. positive trend)?

Statistical model: building blocks



- Model the process that generates the data:
- We want to learn the association of a **single response** variable Y with **one or more predictor** variables $X1, X2, \dots$
- Predictors can be **categorical** (factor, e.g. „warm“ vs „cold“ treatment) or **continuous** (e.g. exact temperature values 11.0°C, 13.9°C, 12.1°C, ...)

Statistical model: building blocks



- Deterministic part:
Prediction model, e.g. mean regression line
- Stochastic part:
The prediction model cannot explain response perfectly, include random error
- Deterministic and stochastic parts both have **parameters** (e.g. effect sizes)

Example: linear regression

Example: linear relationship between age x and size y of adult sea turtles

Deterministic part:

$$\mu(x) = a + b \cdot x$$

Probably a simplification!

Stochastic part:

$$y \sim \text{Normal}(\mu, \sigma)$$

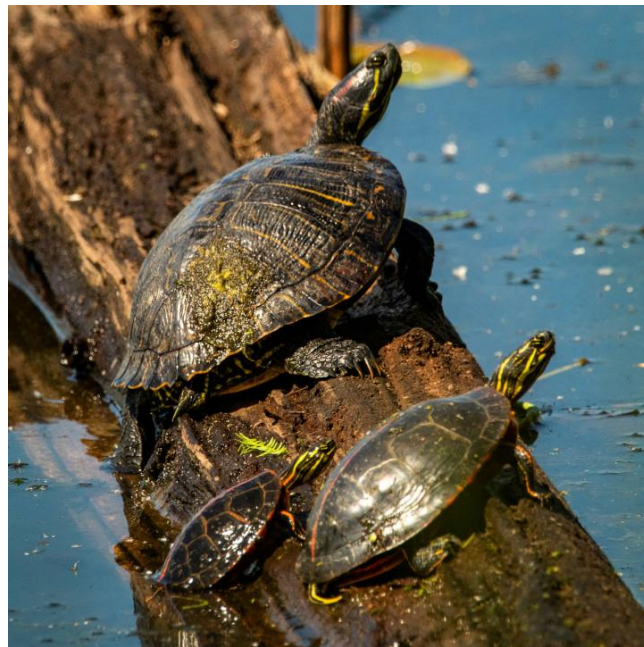
Connects the det. model to the data

Parameters:

a intercept

b slope

σ standard deviation



Example: linear regression

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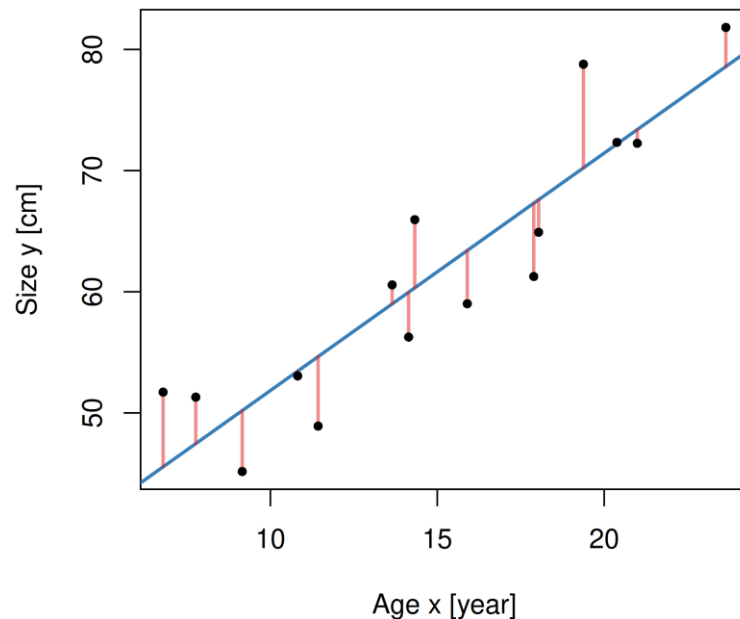
$$y \sim \text{Normal}(\mu, \sigma)$$

Parameters:

a intercept

b slope

σ standard deviation



Example: linear regression

Data: independent observations

$(x_1, y_1), (x_2, y_2) \dots (x_n, y_n)$

Deterministic part:

$$\mu_i = a + b \cdot x_i$$

Stochastic part:

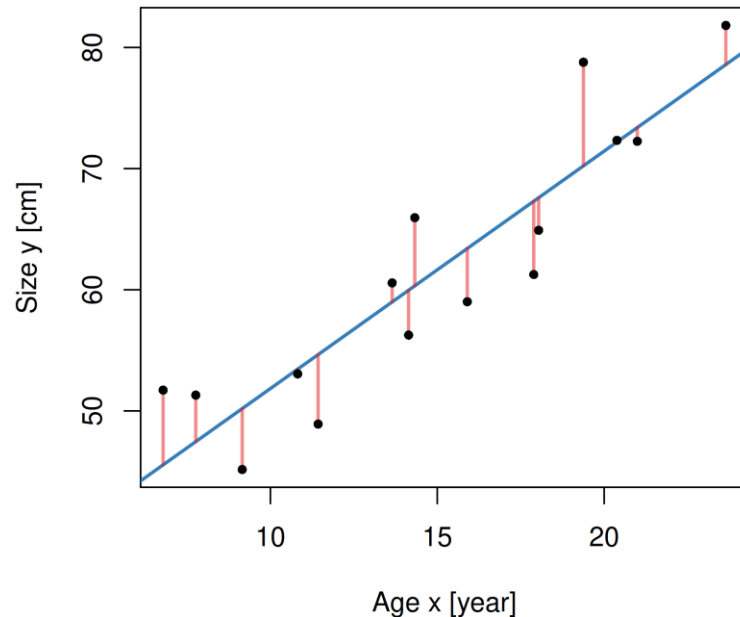
$$y_i \sim \text{Normal}(\mu_i, \sigma)$$

Can be rewritten:

$$y_i = \mu_i + \varepsilon_i$$

$$\varepsilon_i \sim \text{Normal}(0, \sigma)$$

ε_i **residuals** (difference between pred. and obs.)



Example: linear regression

Question:

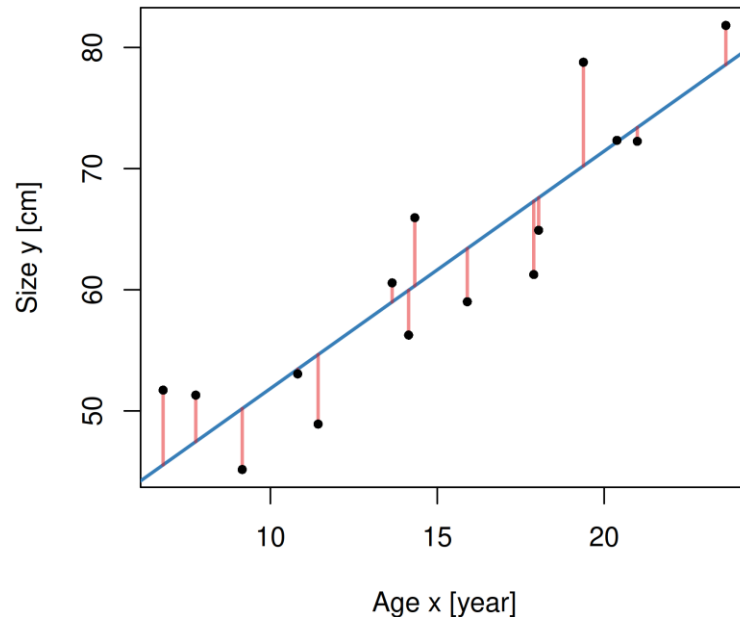
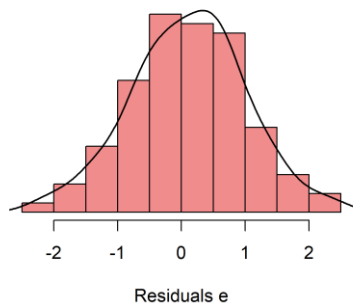
Do datapoints $y_1 \dots y_n$ need to come from a joint normal distribution?

Answer:

No, assumption not about the response values y_i !!!

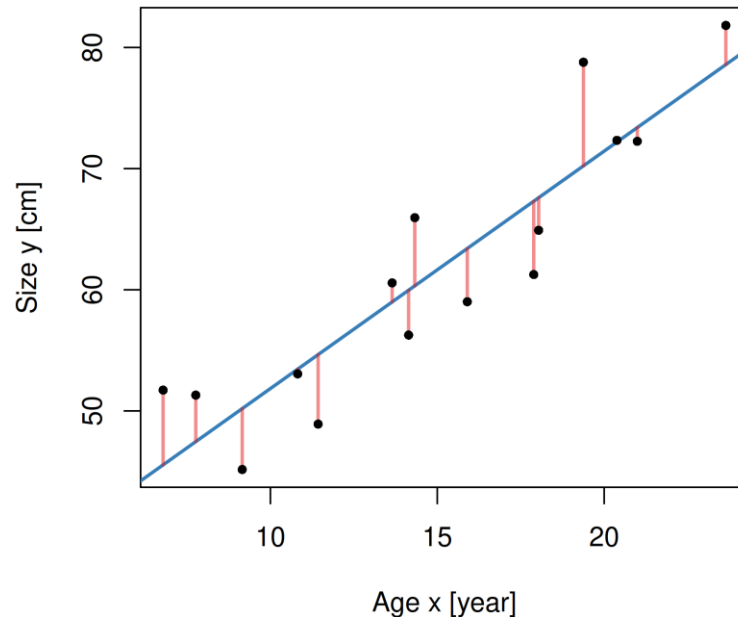
Response y_i has shifting mean: μ_i

Assumption is about the **residuals** ε_i ,
they have a joint zero mean and joint sdev σ



Assumptions in linear regression

- 1. Linearity:** Trend of y follows linear prediction model $\mu(x) = a + b \cdot x$
- 2. Independence** of residuals:
not correlated (e.g. space or time)
- 3. Equal variance** of residuals:
across whole range of x
- 4. Normality** of residuals:
 $\varepsilon \sim \text{Normal}(0, \sigma)$



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Beyond linear models

Generalized linear models, or even nonlinear models allow a wide range of trends

Mixed effects / hierarchical models can account for grouping factors like „plot“

Other distributions with non-constant variance available (e.g. for overdispersion)

Choose other residual distributions to model y (e.g. Poisson for count)

Statistical modeling

There is no such thing as a „Bayesian model“!

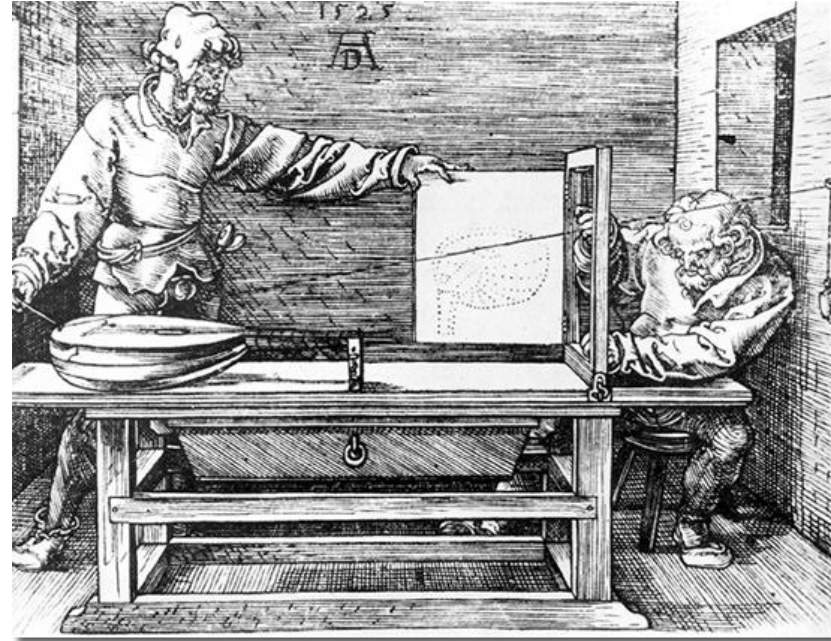
Statistical model:

- Deterministic part
- Stochastic part
- Model assumptions

2 approaches to model fitting / parameter estimation / statements about hypotheses:

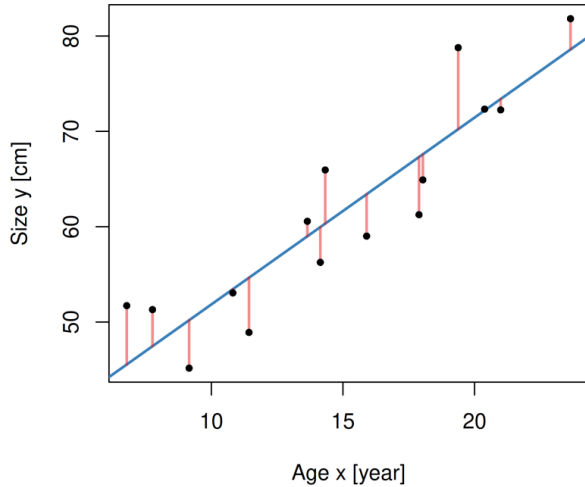
- Frequentist statistics
- Bayesian statistics

They are different in the way model parameters are computed and how their **uncertainty** is treated.



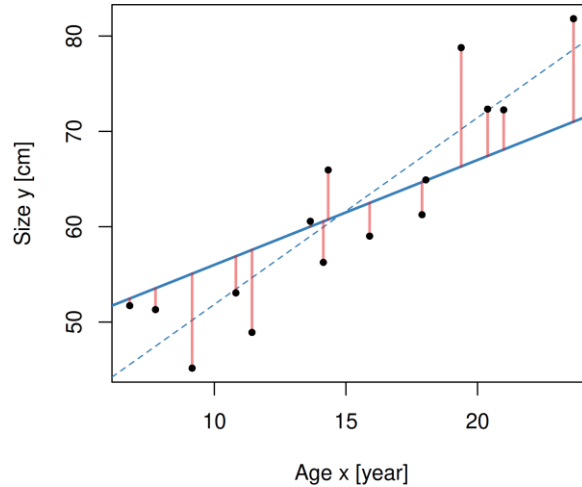
Maximum likelihood estimation

How to estimate parameters?



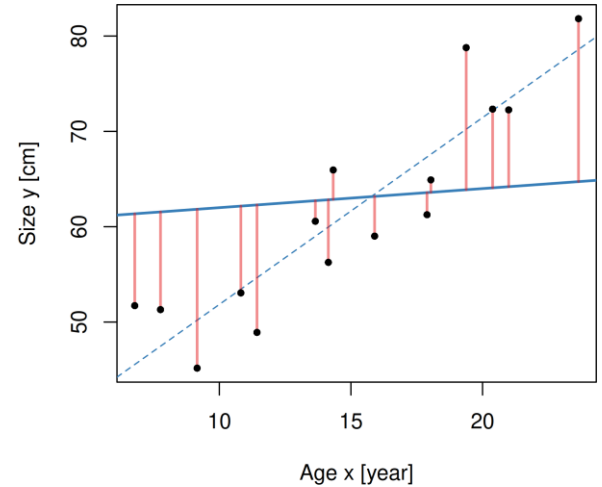
Best model fit:

$$a = 32 \quad b = 2.0$$



Worse fit:

$$a = 45 \quad b = 1.1$$



Really bad fit:

$$a = 60 \quad b = 0.2$$

How to estimate parameters?

- Ordinary **least-squares**

- Find intercept a and slope b that

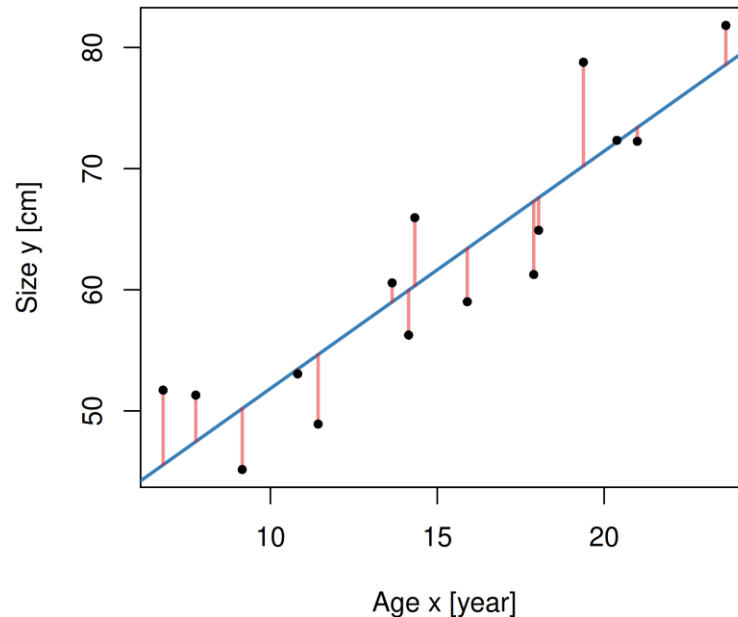
minimize $\sum_{i=1}^n (y_i - \mu_i)^2$ (sum of squares)

- Works perfectly for linear models
- Formulas for intercept and slope(s) available!

- But what about other models (GLM, LMM, ...)?

- Other measure of model fit?

- Stochastic part of the model $\rightarrow$ Probability distribution of datapoints



The likelihood function

Example: survival rate

Statistical model: deterministic part: $\mu = \theta$

stochastic part: $y \sim \text{Binomial}(N, \theta)$

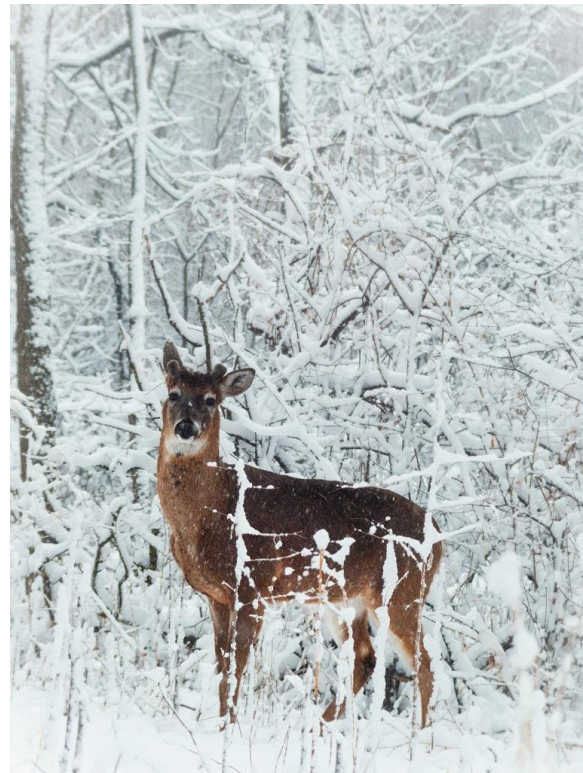
Probability: data unknown, parameters given

- The average survival rate is $\theta = 0.6$
- How many of the 10 individuals will survive the winter?

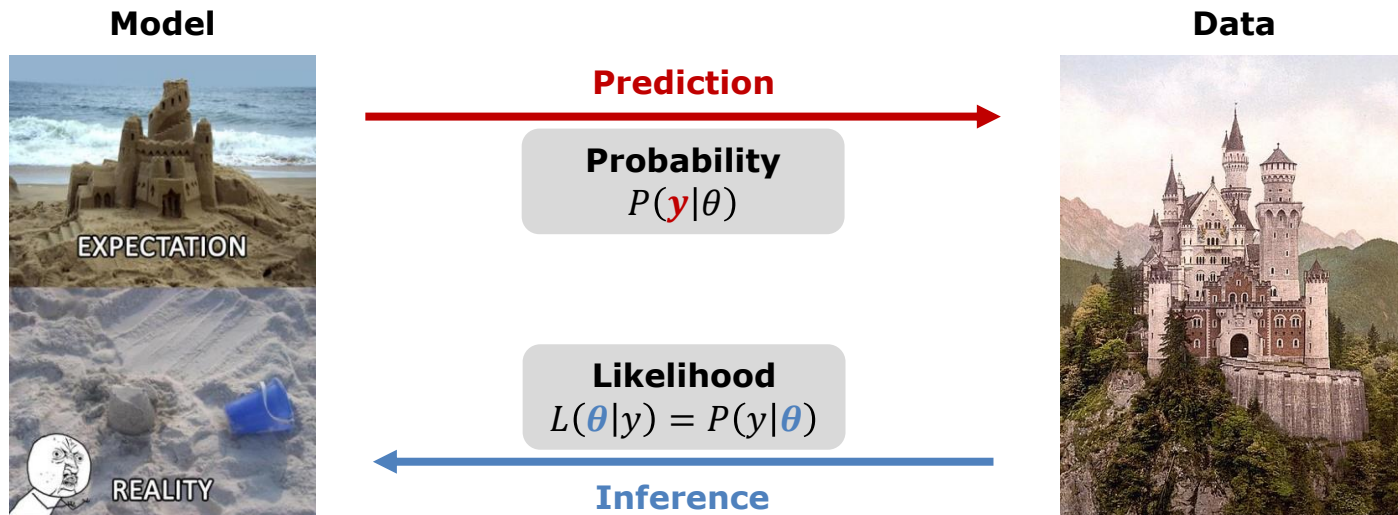
Likelihood: parameters unknown, data given

- Last winter, 6 out of 10 individuals survived
- What is the average survival rate?

→ Likelihood is the **reverse** of probability !



The likelihood function

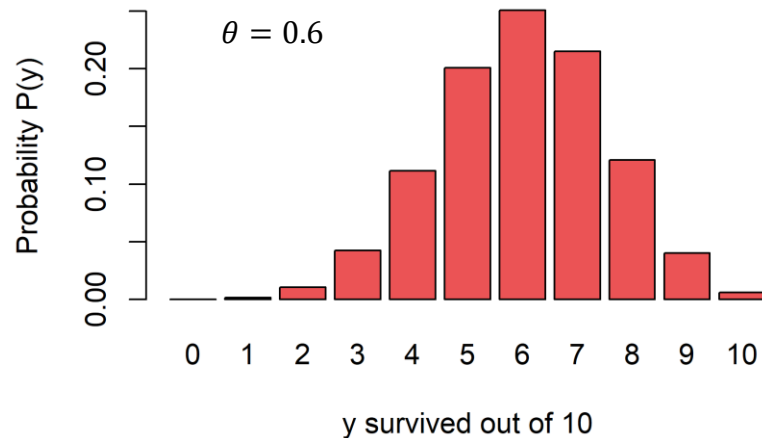


The likelihood function

Probability is function for unknown data

unknown given

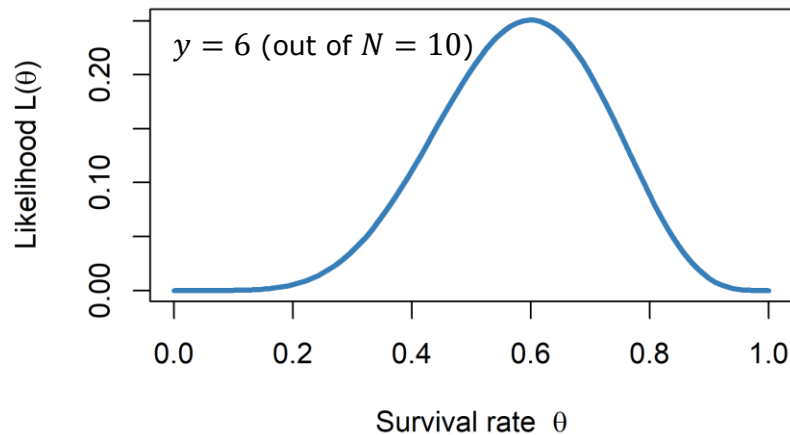
$$\begin{aligned} P(\mathbf{y}|\theta) &= \binom{N}{\mathbf{y}} \theta^{\mathbf{y}} (1 - \theta)^{N-\mathbf{y}} \\ &= \binom{10}{\mathbf{y}} 0.6^{\mathbf{y}} (1 - 0.6)^{10-\mathbf{y}} \end{aligned}$$



Likelihood is function for unknown parameters

unknown given

$$\begin{aligned} L(\theta|\mathbf{y}) &= \binom{N}{\mathbf{y}} \theta^{\mathbf{y}} (1 - \theta)^{N-\mathbf{y}} \\ &= \binom{10}{6} \theta^6 (1 - \theta)^{10-6} \end{aligned}$$



Maximum likelihood estimation

Given:

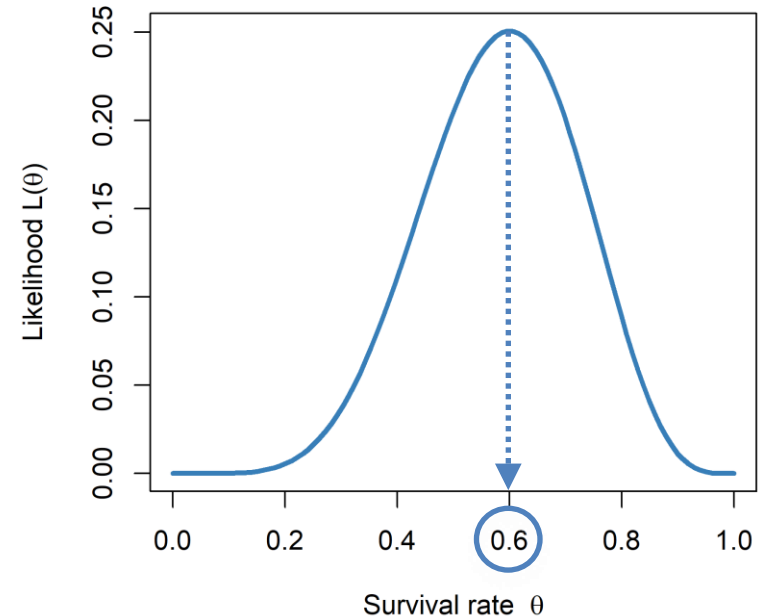
Data y and statistical model

→ Defines likelihood function $L(\theta|y) = p(y|\theta)$

How likely did a parameter value θ produce the observed data?

Find the value for which the likelihood is highest!

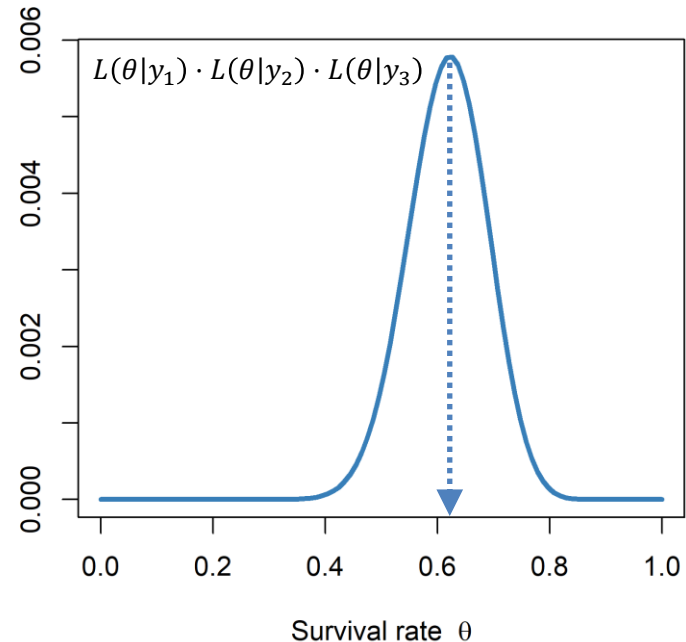
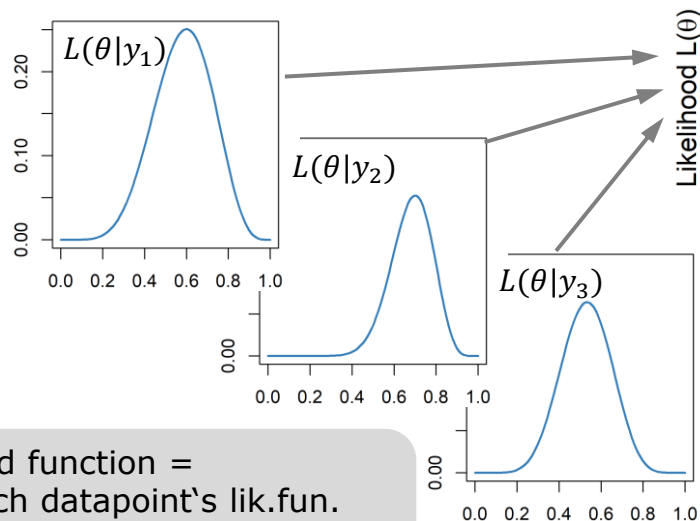
→ We get a **point estimate** θ^*
„Maximum likelihood estimate“



Maximum likelihood estimation

Now: **multiple observations**

Survival: $y_1 = 6/10$, $y_2 = 14/20$, $y_3 = 8/15$



Joint likelihood function =
product of each datapoint's lik.fun.

$$L(\theta|y) = L(\theta|y_1) \cdot L(\theta|y_2) \cdot L(\theta|y_3)$$

Maximum likelihood estimation

Example: linear regression

Deterministic part: $\mu(x) = a + b \cdot x$

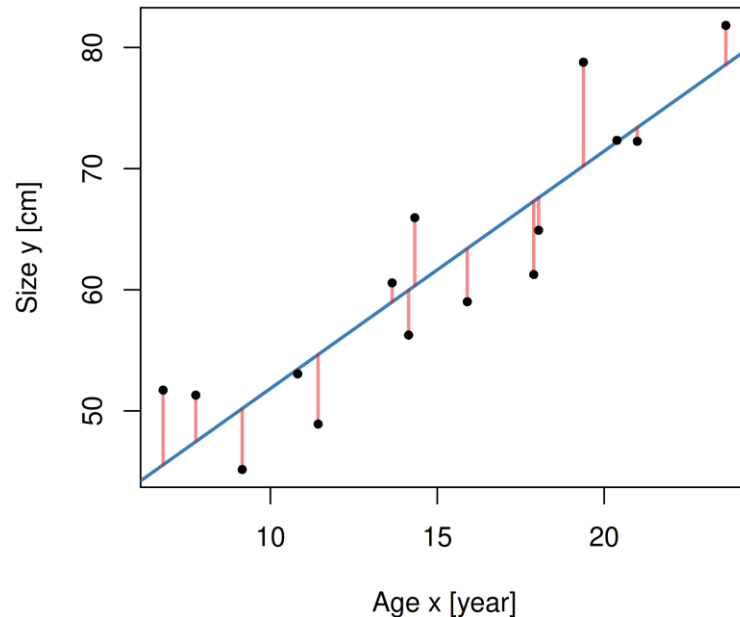
Stochastic part: $y \sim \text{Normal}(\mu, \sigma)$

3 parameters: intercept a , slope b , sdev σ

$$\begin{aligned} L(\mathbf{a}, \mathbf{b}, \boldsymbol{\sigma} | \mathbf{y}) &= p(\mathbf{y} | \mathbf{a}, \mathbf{b}, \boldsymbol{\sigma}) \\ &= p(y_1 | \mathbf{a}, \mathbf{b}, \boldsymbol{\sigma}) \cdot \dots \cdot p(y_n | \mathbf{a}, \mathbf{b}, \boldsymbol{\sigma}) \end{aligned}$$

Now it's getting more complicated:

Find $\mathbf{a}, \mathbf{b}, \boldsymbol{\sigma}$ that maximizes $L(\mathbf{a}, \mathbf{b}, \boldsymbol{\sigma} | \mathbf{y})$



Maximum likelihood estimation

1) Analytical solution: find a mathematical formula for θ

→ Works for linear models with normal distribution

→ But too complicated for most applications

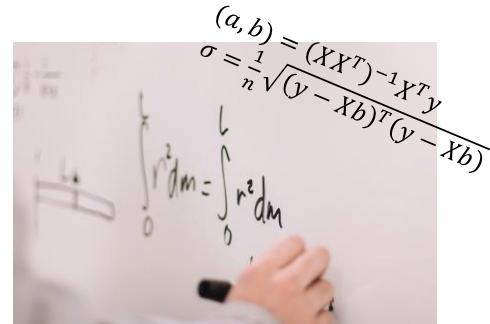
2) Brute force (e.g. grid)

→ Effort grows exponentially with number of parameters

→ Too expensive for most applications

3) Numerical optimization

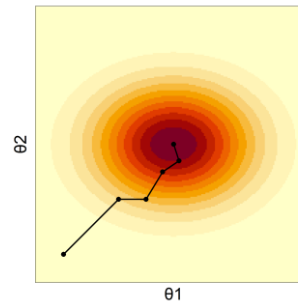
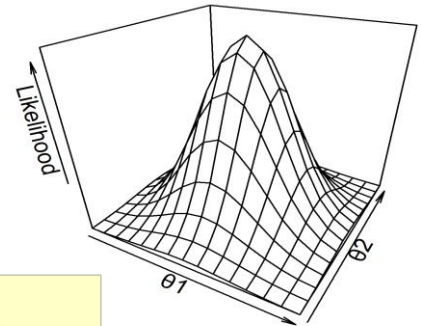
→ Iterative algorithm that tries to improve $L(\theta|y)$ in every step until no further improvement is possible



Hand-drawn equations on a whiteboard:

$$(a, b) = (XX^T)^{-1}X^Ty$$
$$\sigma = \frac{1}{n} \sqrt{(y - Xb)^T (y - Xb)}$$

Below these, there is a small diagram of a rectangle with a vertical line and the label $\int_0^L r^2 dm = \int_0^L r^2 dm$.



Beyond point estimates ?

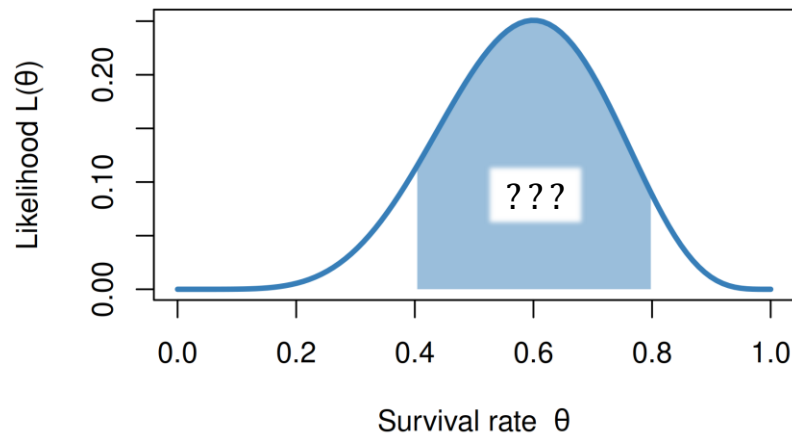
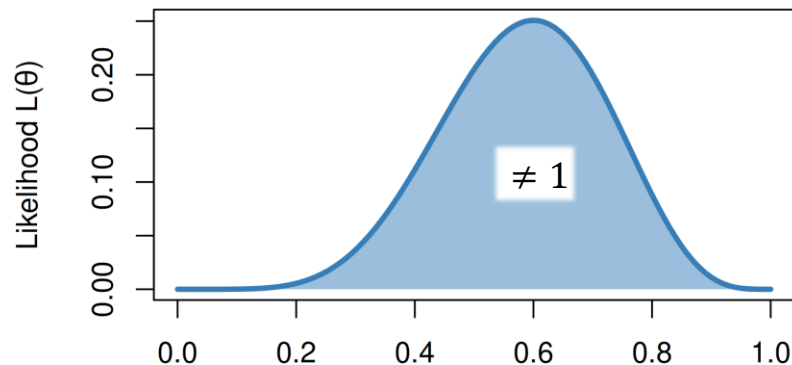
Why can't we use the likelihood for probability statements on the parameters ?

$L(\theta|y)$ is not a probability density function for parameters θ !

$\int L(\theta|y) \neq 1$ (area under the curve)

E.g. $\int_{0.4}^{0.8} L(\theta|y)$ is a meaningless value.
It does **not** describe $P(0.4 < \theta < 0.8)$!

But likelihood tells us that, e.g., survival rate of 0.3 is less likely than 0.5. Can we use that?



Beyond point estimates ?

$$L_{new}(\theta|y) = \frac{L(\theta|y)}{c} \quad \text{scale by constant } c = \int L(\theta|y)$$

$$\int L_{new}(\theta|y) = 1 \quad (\text{area under the curve})$$

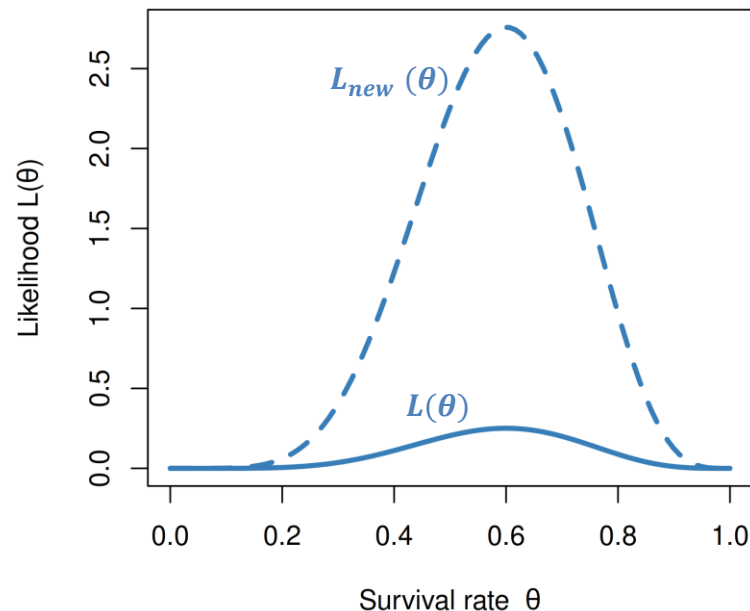
Probability statements would be possible!

$$\text{E.g. } \int_{0.4}^{0.8} L_{new}(\theta|y) = P(0.4 < \theta < 0.8)$$

But we arrived at the same problem:

Can't compute the integral $c = \int L(\theta|y)$

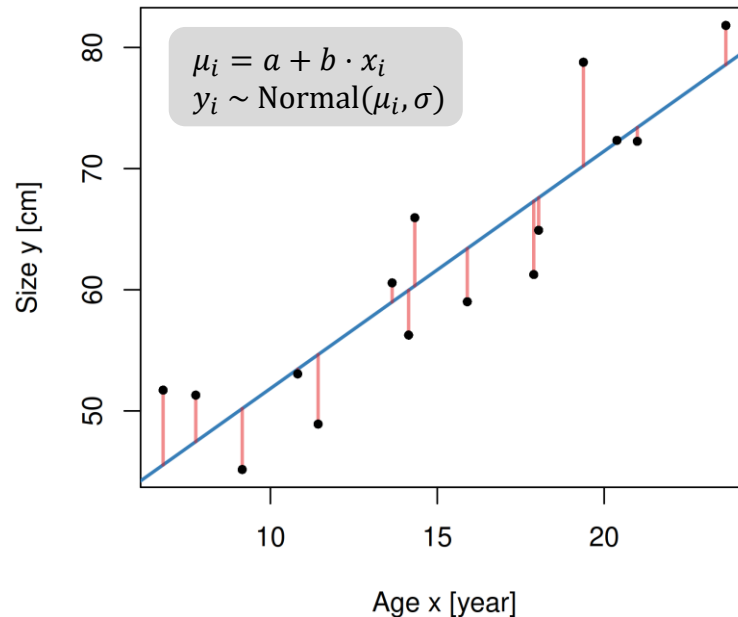
It's not practical. Solution in next lecture!



Summary

Summary MLE

- Every statistical model has a likelihood function, defined by distribution of the stochastic part, that connects deterministic part to data (prob of the data, given a fixed parameter)
 - Find model parameters such that observed data is most likely
 - Maximum likelihood estimation → point estimates
 - Does not allow probability statements about the model parameters $P(\theta|y)$
- The frequentist „short cut“:
Null hypothesis significance testing (NHST)



Further reading

Fieberg, J. (2024). Statistics 4 Ecologists. <https://statistics4ecologists-v2.netlify.app/> [Chapters 1,9,10]

Inchausti, P. (2023). Statistical Modeling With R: a dual frequentist and Bayesian approach for life scientists. *Oxford University Press*. [Chapter 3]

Kery, M. & Kellner, F. (2024): Applied Statistical Modelling for Ecologists. *Elsevier*. [Chapters 1,2]

Essington, T. (2021). Introduction to Quantitative Ecology. *Oxford University Press*. [Chapter 8]

Warton, D. (2022). Eco-Stats: Data Analysis in Ecology. *Springer (Methods in Statistical Ecology)*. [Chapter 1]